

**SIXTH SEMESTER (CUCBCSS—UG) DEGREE [SPECIAL] EXAMINATION
MARCH 2021**

Mathematics

MAT 6B 13 (E03)—C PROGRAMMING FOR MATHEMATICAL COMPUTING

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

Maximum : 20 Marks

INSTRUCTIONS TO THE CANDIDATE

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MAT 6B 13 (E03) – C PROGRAMMING FOR MATHEMATICAL COMPUTING

(Multiple Choice Questions for SDE Candidates)

1. # Define is a :

- (A) Statement.
- (B) Preprocessor compiler directive.
- (C) Function statement.
- (D) None.

2. Which one among the following statements is true ?

- (A) Every C program ends with an END word.
- (B) Main () is where the program stops its execution.
- (C) A line in a program may have more than one statement.
- (D) All variables in C must be declared for their types before they are used in the program.

3. The modulus operator % can be used :

- (A) Only for floating point data.
- (B) Integer data.
- (C) Exponential functions.
- (D) None.

4. What will be the output of the program :

```
#include <stdio.h>

int main( )
{
    int y =12 ;
    const int x = y ;
    printf( " %d\n", x);
    return 0;
}
```

- (A) 12.
- (B) Garbage value.
- (C) Error.
- (D) None.

5. What will be printed when the sample code below is printed ?

```
int x =0;
for( x=1; x<4; x++);
printf("x= %d\n", x);
```

(A) 0.

(B) 1.

(C) 3.

(D) 4.

6. An integer constant in *c* must have :

(A) At least one digit.

(B) Commas and blanks.

(C) Decimal points.

(D) None.

7.

```
int i =3;
```

```
switch =4;
```

```
{
```

```
default:
```

```
;
```

```
case 2:
```

```
i+= 4;
```

```
if (i == 8)
```

```
{
```

```
    i++;
```

```
if (i == 9) break ;
```

```
    i +=1;
```

```
}
```

```
i- = 3 ;
```

```
break;
```

```
case 8
```

```
    i+= 4
```

```
break;
```

```
}
```

```
printf( "i = %d\n",i)
```

Turn over

The output of the code is” :

- (A) i = 4.
- (B) i = 6.
- (C) i = 9.
- (D) i = 8.

8. When applied to a variable, what does the unary operator yield ?

- (A) The variable's address.
- (B) The variable's right value.
- (C) The variable's binary form.
- (D) The variables Value.

9. Text enclosed in a pair of quotation marks is a data type :

- (A) Integer.
- (B) Long.
- (C) String.
- (D) Variable.

10. A memory location with some data that will not change is a :

- (A) Constant .
- (B) Variable.
- (C) String.
- (D) Integer.

11. The keyword used to declare a variable is :

- (A) Const.
- (B) Var.
- (C) String.
- (D) Dim.

12. To declare more than one variable on the same line, separate the variables with :

- (A) Commas.
- (B) Colons.
- (C) Pipes.
- (D) Semicolons.

13. What is accomplished by the assignment statement "Hourlyplan = txtPay.Text" ?
- (A) The value in the variable Hourly Plan is copied into the txtPay.Text text box.
 - (B) The value in the variable Hourly Plan is compared to the value in txtPay .
 - (C) The value entered in the txtPay text box is copied into the variable Hourly plan.
 - (D) The value entered in the txtPay text box is compared to the value in the variable \ Hourly plan.
14. An expression can be a :
- (A) Constant.
 - (B) Variable.
 - (C) Combination of constants, variables, and arithmetic operations that result in a value.
 - (D) All of the above.
15. A string literal must be enclosed in :
- (A) Quotation marks (").
 - (B) Single quotes (').
 - (C) Pound signs (#).
 - (D) Exclamation points(!).
16. A memory location with data that will change is recognized as :
- (A) Variable.
 - (B) Constant.
 - (C) Keyword.
 - (D) None.
17. The format specification for printing an integer number with minimum field width w is :
- (A) % wf.
 - (B) % ld.
 - (C) % wd.
 - (D) % wf.
18. The operator ? : is :
- (A) A multi-way decision operator.
 - (B) Two-way conditional operator.
 - (C) Loop operator.
 - (D) None.

Turn over

19. The while statement is an :

- (A) Exit controlled loop statement.
- (B) Entry controlled loop statement.
- (C) Both (A) and (B).
- (D) None.

20. Which of the following statement is correct for For loop ?

- (A) Entry controlled loop.
- (B) Nesting is allowed.
- (C) The starting value of the control variable must be less than its ending value.
- (D) Both A and B above.

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MARCH 2021**

Mathematics

MAT 6B 13 (E03)—C PROGRAMMING FOR MATHEMATICAL COMPUTING

Time : Three Hours

Maximum : 80 Marks

Section A

Answer all questions.

Each question carries 1 mark.

1. Which symbol is used to denote a pre-processor statement ?
2. What is an executable object code ?
3. What are the different categories of operators in C ?
4. What are the logical operators ? Give meaning of each.
5. Write the syntax of conditional operators.
6. Distinguish between = and == operators in C.
7. What is the new line character in C ?
8. What is the header file containing getch() function.
9. Write the syntax of the **while** statement in C.
10. What is the use of clrscr () in C ?
11. What is debugging a program ?
12. Name the inventors of the programming language C.

(12 × 1 = 12 marks)

Section B

*Answer at least **eight** questions.*

Each question carries 3 marks.

All questions can be attended.

Overall Ceiling 24.

13. What are the steps involved in executing a C program ?
14. What are the character categories in C ?
15. Write a brief note on keywords.

Turn over

16. Give any two branching statements with syntax.
17. Write the syntax and draw the flow chart of the do..... while statement.
18. Write a short note on header files.
19. Give any two drawback of switch statement.
20. Write a short note on continue statement.
21. Specify escape sequences and their purpose in C.
22. What are the effects of the formats %d, %f, %lf and %g in printf function.
23. What is meant by function prototype ?
24. Specify any two string handling functions and their actions.

(8 × 3 = 24 marks)

Section C

*Answer at least five questions.
Each question carries 6 marks.
All questions can be attended.
Overall Ceiling 30.*

25. Explain in detail any three input statements with suitable example.
26. Discuss about different types of errors in C.
27. Write a program to find the roots of a quadratic equation with a provision for inputting coefficients of the equation.
28. Compare break and goto statements.
29. Write a program to print the following output using for loop :

```

      1
    2  2
  3  3  3
4  4  4  4
5  5  5  5  5

```

30. What is an array ? Explain the need for array variable.
31. Explain multiple branching statements in C.
32. Explain user defined functions in C.
33. Write a note on reading strings from terminal with suitable example.

(5 × 6 = 30 marks)

Section D

*Answer any **one** question.*

The question carries 14 marks.

34. Explain, in detail with syntax, flow chart, and examples, different looping statement in C.
35. a) Explain the handling of multidimensional arrays with syntax and suitable example.
b) Briefly explain different categories of functions.
36. a) Explain the working of any two string handling functions with syntax.
b) Write a program to copy one string to another and count the number of characters copied, (without using string handling functions).

(1 × 14 = 14 marks)

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